



Financial networks: a retrospective view to looking ahead

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ABSTRACT

This study defines the cognitive structure of financial networks research by looking at the terms scholars have used since 1990. Applying co-word analysis over a sample of 680 records extracted from Scopus database, the findings reveal that the literature on financial networks is made up of six thematic clusters, these are: *Dynamics of financial networks*, *Complex banking networks*, *Risk and contagion in financial networks*, *Topology and structure of financial networks*, *Main methodologies and techniques* and *Post-crisis responses*. In addition to laying the groundwork for understanding financial networks through a retrospective view, this study shows how bibliometrics (especially co-words analysis) can contribute to reducing conceptual ambiguity in emerging research fields, such as financial networks research, and open doors for future research.

Keywords: Financial networks; Co-words analysis; Intellectual structure

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Redes financieras: una visión retrospectiva para mirar al futuro

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RESUMEN

Este estudio define la estructura cognitiva de la investigación sobre las redes financieras examinando los términos que los académicos han utilizado desde 1990. Aplicando un análisis de palabras conjuntas sobre una muestra de 680 registros extraídos de la base de datos Scopus, los resultados revelan que la literatura sobre redes financieras está formada por seis grupos temáticos, que son: Dinámica de las redes financieras, Redes bancarias complejas, Riesgo y contagio en las redes financieras, Topología y estructura de las redes financieras, Principales metodologías y técnicas y Respuestas posteriores a la crisis. Además de sentar las bases para la comprensión de las redes financieras a través de una visión retrospectiva, este estudio muestra cómo la bibliometría (especialmente el análisis de co-palabras) puede contribuir a reducir la ambigüedad conceptual en los campos de investigación emergentes, como la investigación de las redes financieras, y abrir puertas para futuras investigaciones.

Palabras clave: Redes financieras; Análisis de co-palabras; Estructura intelectual

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1. Introduction

Financial networks are models employed to characterize and analyze the dynamics and interrelationships of financial institutions and markets which are characterized by a high degree of complexity and uncertainty. Their main function is to represent the conjunct interconnections of a system composed of vertices (financial agents) and edges (financial interconnections which can be financial claims, debt, flows of money, etc.) (Berdson, 2018). Therefore, financial networks are capable of representing the full picture of a financial system and extract insights from quantitative techniques, where traditional economic, financial and econometric models are not capable of.

Financial networks came hand in hand with the revolution of network science in the 1990's, when it became clear that the biggest challenge of science is to represent the conjunct behavior of its subjects of study, as the different parts of them are usually non-separable and need a full representation to be studied (Newman et al., 2006). However, there certainly is a breaking point in the financial networks literature, which is the 2007-2009 financial crisis. It is during the explosion of this crisis when it became clear that the economic, econometric and financial models that were being used in production failed to detect such a shock, and did not give any warning signs of what was occurring behind the scenes (Bordo and Meissner, 2016).

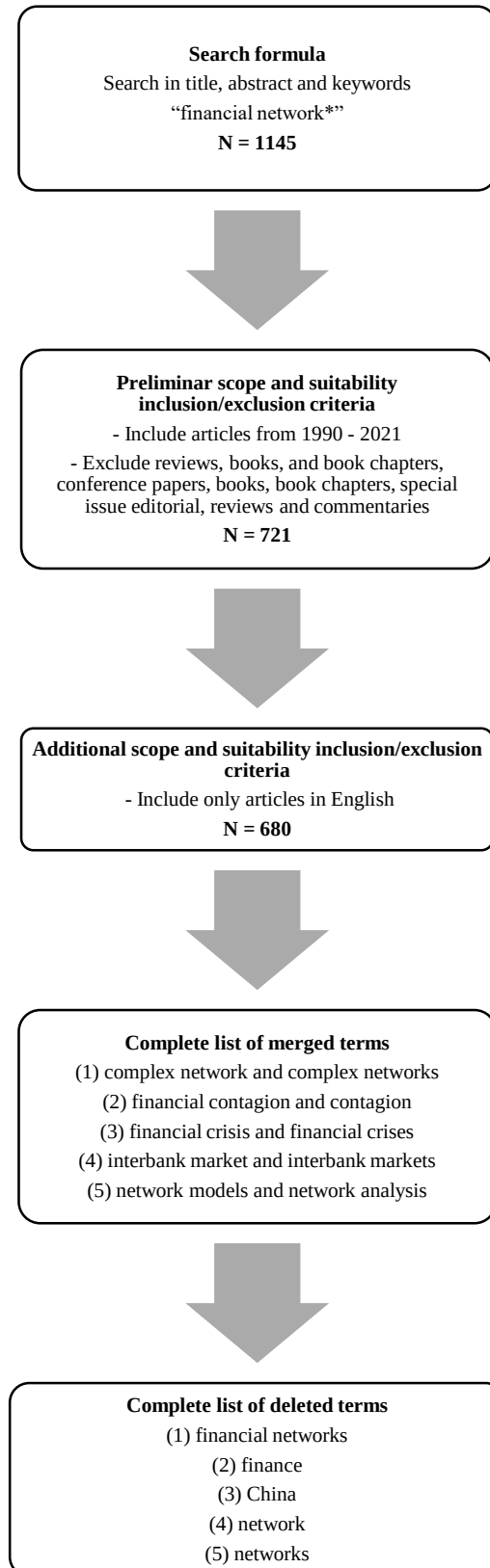
It is from that point that financial authorities started to switch their interests to financial network models to engross stress tests and macro-econometric models to include the systemic components, and this path led directly to financial network models (Battinson and Martinez-Jaramillo, 2018). Thus, there is an explosion in the research interest in this field from that point, as this paper reveals. However, it is crucial to determine the path of the interest in financial networks and in which point is situated nowadays, as it has been widely concluded that financial crises are a recurrent phenomenon (Reinhart and Rogoff, 2009). Therefore, if this explosion in the literature of financial networks is only due to recent events, and is not maintained in the future, it is very likely that the world will not be prepared for the next financial crisis because stress tests and quantitative models will be underdeveloped. Thus, this research aims to analyse the cognitive structure of the financial networks in order to make a retrospective analysis to look into the future (cf. Donthu et al., 2021; Zupic and Čater, 2015).

Therefore, in this paper the authors address this by conducting a bibliometric analysis (specifically a co-words analysis) of the literature in financial networks and revealing its main trends and patterns since its origin as a discipline in 1990. To this, we employ the literature of the largest scientific database in the world, SCOPUS. The rest of the paper structure is as follows. Section 2 explains the materials and methods employed. Section 3 discusses the results through the lens of the literature. Finally, section 4 concludes.

2. Methodology

For this study we selected the SCOPUS database as the data source since it has been recognized as the most inclusive academic database (Mongeon and Paul-Hus, 2016). We used the word “financial network*” and excluded all records that do not meet the inclusion criteria. We considered all records published without considering the publication year 2022 and live aside conference papers, books, book chapters, special issue editorial, reviews and commentaries (Terán-Yépez et al., 2021). We also applied as inclusion criteria that we will only focus on studies published in English (Figure 1 show the results refinement procedure). As a final sample, we obtained 680 studies. Then, we download all records and incorporate them in the VOSviewer software (Van Eck and Waltman, 2010; Van Eck and Waltman, 2014). Once in the software we refine and pre-process the manuscripts retrieved by using a Thesaurus file. Thus, for example we merge the some synonym terms such as complex networks and networks and financial contagion and contagion, or eliminate very generic terms such as finance or networks and the words used as keywords (i.e., financial networks) to perform the search. Finally, we run the co-words analysis with VOSviewer software (Van Eck and Waltman, 2010).

Figure 1 Methodology flowchart



3. Results and Discussion

This study extrapolated 680 records. The first set of works in financial networks as a discipline started in the decade of 1990, motivated by advances in network science as a new way to conduct research understanding the subjects of study as a complex system with multiple dynamics that cannot be divided. During this decade, the development of network science was primarily done in fields such as physics, electrical engineering and sociology. There were only 19 papers dedicated to financial networks during this decade.

However, considering the number of records generated during each decade, as Table 1 shows, it is possible to notice an exponential increase in the number of studies dedicated to financial networks, with 65 articles published in the decade of 2001-2010 and 596 published in the decade of 2011-2021, which translate to incrementation rates of 342% and 916% with respect to the previous decade.

If one further compares the incrementation rate of 2011-2021 with the one of 1990-2000, it is possible to see that the incrementation rate is 3136% in this case. Furthermore, the number of records during the decade of 2011-2021 rises to 596, in comparison with 65 and 19 of the decades of 2001-2021 and 1990-2000.

What explains this incredible rise in the number of publications in the field, apart from the development of network science as a discipline, is that it was clear that macroeconomic, monetary and financial models that researchers and practitioners were using failed to provide warning signs about the financial crisis of 2007-2009. Thus, since the financial crisis there is a generalized agreement in the literature that there is a necessity for accounting to financial interconnections, systemic components and its relationship with the economy and the financial system as a whole.

Table 1 Number of records for decade and incrementation rates of publications

Publishing years	Records No.	% Incrementation rate
1990 - 2000	19	
2001 - 2010	65	242%
2011 - 2021	596	817%

As the advances in network theory were firstly done in fields such as physics, it is not a surprise that a physics journal dominates the field of financial networks. This is directly reflected in Table 2, which reflects the list of publications, citations, and the ratio of citations per publication of the most active journals in the field. *Physica A: Statistical Mechanics and its Applications* dominates the field with 45 publications, 632 citations, and 14 citations per document on average, followed by the *Journal of Economic Dynamics and Control* (23 publications, 811 citations and 35 citations per document on average) and the *Journal of Financial Stability* (22 publications, 442 citations, and 20 citations per document on average).

Table 3 reflects the most active authors in the financial networks field. There is a clear liderage by Steffano Battinson, with 18 publications, 1145 citations, and 64 citations per document on average. In fact, this author has more citations than the following three authors altogether. The second most active author is Anna Nagurney, who has 12 publications in the field, 222 citations and 19 citations per document on average, followed by Andreaa Minca, with 11 documents, 248 citations and 23 citations per document on average.

Table 2 List of most active journals in financial networks by number of publications, number of citations, and citations per document

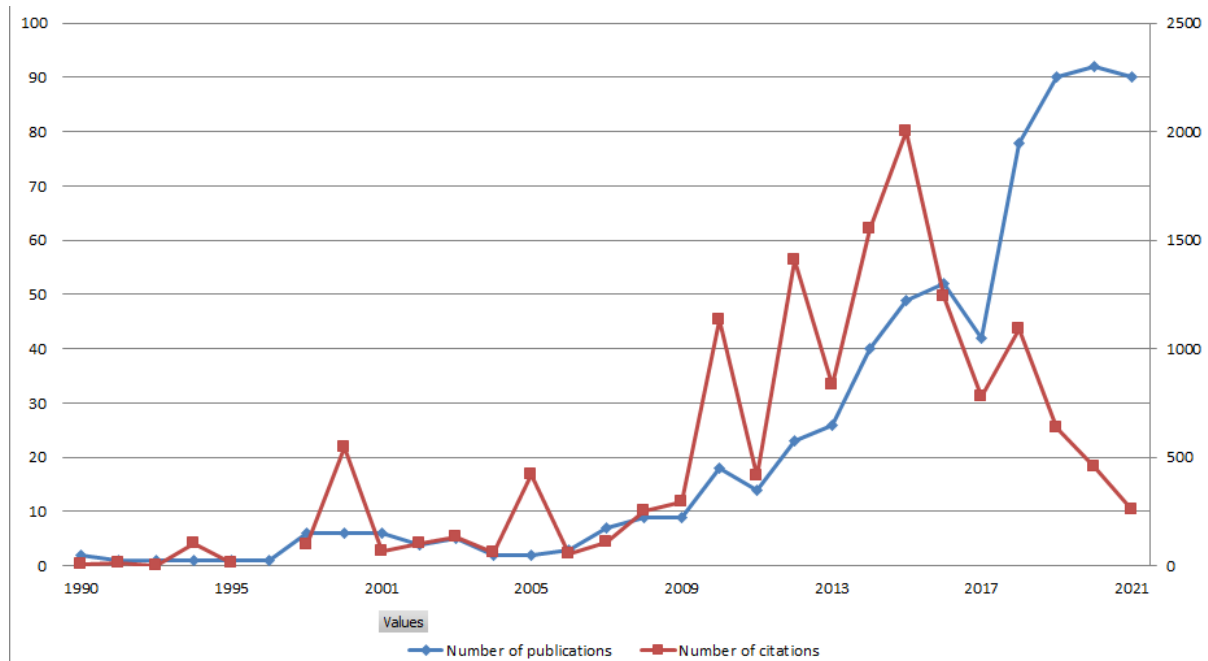
A	Journal	C	C/A
46	Physica A: Statistical Mechanics and its Applications	632	14
23	Journal of Economic Dynamics and Control	811	35
22	Journal of Financial Stability	442	20
16	Scientific Reports	760	48
15	PLoS ONE	316	21
15	Journal of Banking and Finance	759	51
12	Quantitative Finance	331	28
11	Complexity	100	9
11	Journal of Network Theory in Finance	46	4
10	Computational Economics	66	7
10	Journal of Statistical Mechanics: Theory and Experiment	89	9
10	SIAM Journal on Financial Mathematics	66	7

Table 3 List of more active authors in the financial networks literature by number of publications, number of citations and citations per document

A	Author	C	C/A
18	Battiston, S.	1145	64
12	Nagurney, A.	222	19
11	Minca, A.	248	23
11	Silva, T.C.	265	24
10	Caccioli, F.	577	58
10	Tabak, B.M.	200	20
10	Turner, S.	183	18

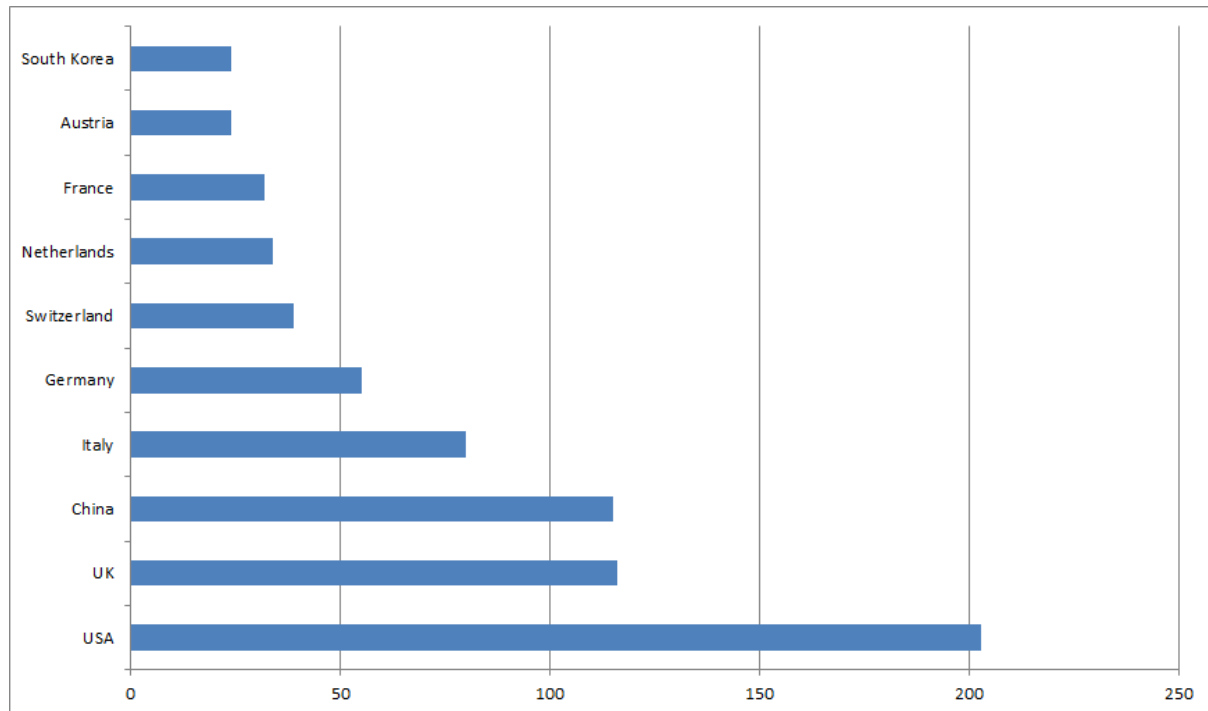
Considering the dynamic evolution of the field, it is possible to see in Figure 2 that there has been an explosion of this literature since the financial crisis of 2009. Since 2009, the evolution of both the number of publications (blue line) and the number of citations (red line) has evolved exponentially. However, from 2014-2015, there has been a decoupling of this covariance, as since the number of publications has continued to increase exponentially, the number of citations has decreased, reaching the pre-financial crisis levels in 2021. Presumably, this is due to the passing of the financial crisis, and because of it, of researchers moving their interest to new phenomena such as COVID-19.

Figure 2 Graphical representation of the number of publications and citations of the financial networks literature



As with many other fields, the literature is dominated by the Anglo-Saxon community, as Figure 3 reflects. The USA has the biggest number of publications with more than 200, followed by the UK with more than 100, and by People's Republic of China with a similar number. From there, there are a set of European countries that are pretty active in the field, including Italy, Switzerland, the Netherlands and France.

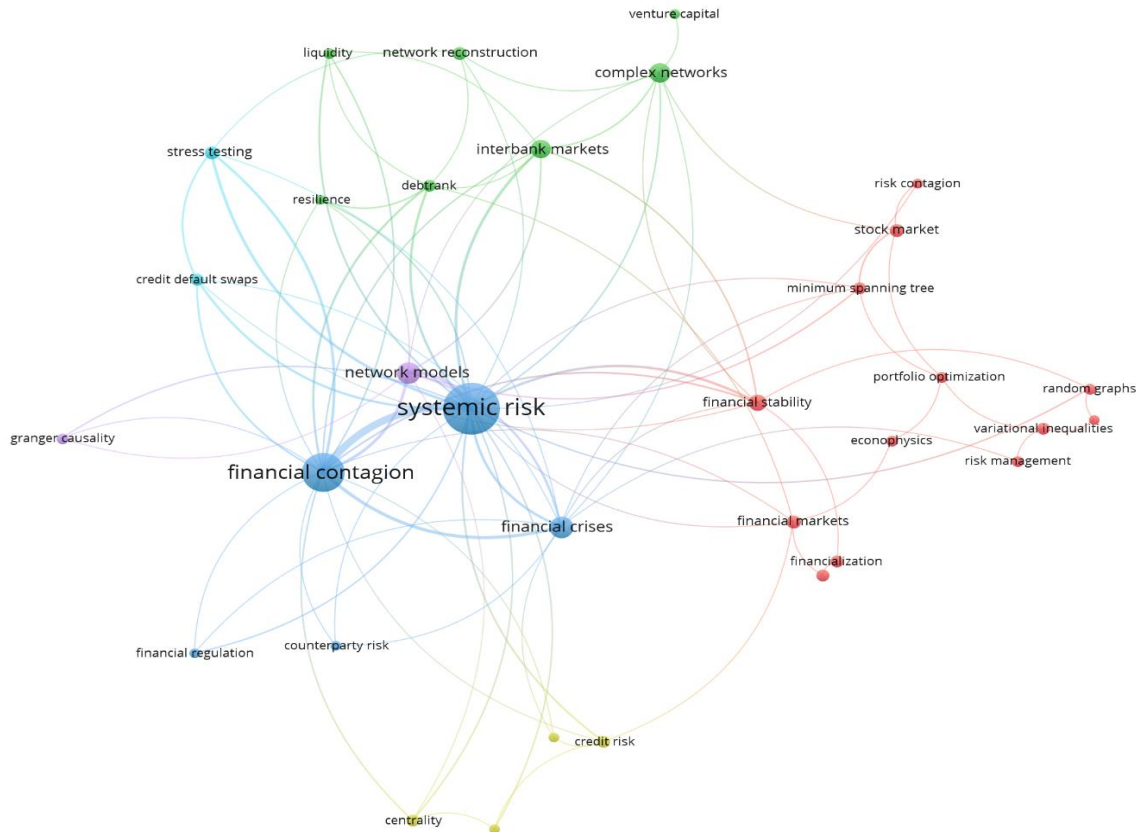
Figure 3 Graphical representation of the number of publications per country of the financial networks literature.



Eventually, in order to analyse the intellectual structure we provide a network of keyword co-occurrence (Donthu et al., 2021; Zupic and Čater, 2015) from all the records extracted from SCOPUS. The graphical representation of this network allows us to infer six clusters that dominate the field. This

graphical representation is provided in Figure 4, and we provide a description of each cluster to reveal a picture of the literature on this field.

Figure 4 Clusters of the financial networks literature



Cluster 1 (red coloured): *Dynamics of financial networks* (13 items). This cluster groups the most common fields of financial networks. Financial stability (Nier et al., 2007; Battinson and Martinez-Jaramillo, 2018; Berdsen et al., 2018; Acemoglu et al., 2015) financial markets (Aldasoro and Alves, 2018; Luu et al., 2021; León et al., 2018) stock market (Boginski et al., 2005; Lee et al., 2019) and econophysics (Chatterjee and Chakrabarti, 2007; Emmert-Streib and Dehmer, 2010) the principal domains of financial networks, and where financial network modelling is mainly developed and applied. Within these fields, financial networks have applications to several subdomains.

Cluster 2 (green coloured): *Complex banking networks* (7 items). This is the second cluster that attaches the complexity of financial markets and the importance of network models to account for that complexity. Its main components are interbank markets where complexity appears (Aldasoro y Alves, 2018; Luu et al., 2021; León et al., 2018) networks which try to model complexity (Ferreira et al., 2020; Bougheas and Kirman, 2015), and network reconstruction (Anand et al., 2018).

Cluster 3 (blue coloured): *Risk and contagion in financial networks* (5 items). This is the main cluster and accounts for the importance of network models to financial stability and systemic risk. Systemic risk is the key component (Battinson et al., 2012; Acemoglu et al., 2015; Roukny et al., 2018), the risk that a financial shock spreads to the financial system due to interconnectedness and affects the financial system as a whole. This is by far the most important concept for the application of financial networks. Financial contagion (Gai and Kapadia, 2010; Glasserman and Young, 2015) is the phenomenon that characterizes systemic risk, that is, that a shock in the USA may affect the financial system of Sweden because of the interconnection of the financial system. Finally, financial crises (Reinhart and Rogoff, 2009) arise because of financial contagion and systemic risks.

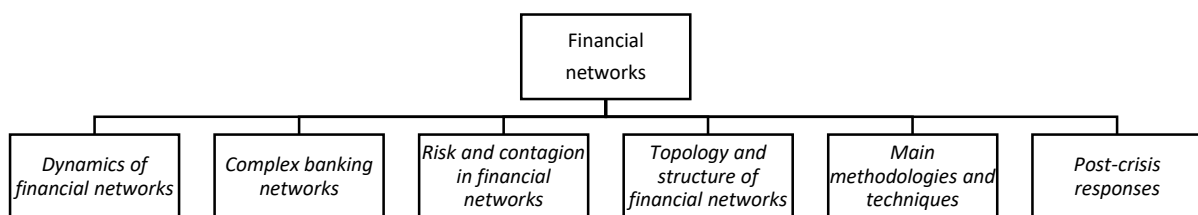
Cluster 4 (yellow coloured): *Topology and structure of financial networks* (4 items). This cluster includes important components of the topology and structure of the financial system modelled by financial networks, such as the centrality of a node.

Cluster 5 (violet coloured): *Main methodologies and techniques* (2 items). The cluster includes the main methodologies for financial networks (network models) and techniques relevant to dynamic networks, such as granger causality (Nier et al., 2007).

Cluster 6 (light blue coloured): *Post-crisis responses* (2 items). An immediate response to the 2007-2009 financial crisis. Credit Default Swaps (CDS) which were some of the main derivatives that originated the crisis, and stress testing (Battinson and Martinez-Jaramillo, 2018; Markose et al., 2012), which demonstrated not being enough to predict the crisis and its advancing in its systemic window by employing network models.

Figure 5 shows the six distinctive dimensions of financial networks.

Figure 5 The cognitive structure of financial networks



4. Conclusion

Based on a co-words analysis, this study delimited the fundamental financial network dimensions resulting from 31 years of research. As such, the main contribution of our findings is that we were able to identify the cognitive structure of financial networks that mirrors the collective scientific understanding of the topic (see Figure 5) which may open up a number of future avenues of research. Our results reveal that financial networks research is made up of six thematic clusters, which are: Dynamics of financial networks, Complex banking networks, Risk and contagion in financial networks, Topology and structure of financial networks, Main methodologies and techniques and Post-crisis responses. Each cluster includes a set of specific terms conveying the distinctiveness of these dimensions for financial networks research. Thus, this research has important implications for the literature, as it has contributed to unravelling the structure of the literature on financial networks by exploiting the distinctive features of co-word analysis and, therefore, to reach a fine-grained perspective on this phenomenon.

According to the words used by the authors to describe their studies and based on the joint word analysis, we propose the following four topics as emerging research streams: (1) stress testing, (2) risk contagion, (3), financial markets, and (4) granger causality. Looking ahead, financial networks scholars need to situate stress testing, risk contagion, and financial markets at the centre of attention. Firstly, since stress testing has become a relevant tool used by financial authorities to determine the stability capacity of an institution or banking system, stress testing frameworks should be further develop by integrating liquidity and solvency interactions to strengthen their macroprudential dimension (Battiston and Martinez-Jaramillo, 2018). Second, as the global financial market has become extremely interconnected as it demonstrates strong nonlinear contagion in times of crisis, and because an increase in financial networks (integration of a greater number of entities) also brings with it an increase in the probability and extent of financial contagion, financial network scholars should propose new systemic risk indicators that can forecast the occurrence and trace the development of financial crises in order to try to control the risk of contagion within financial networks. Finally, given the complexity and uniqueness of financial networks, we believe that that it is necessary to deepen some methodological approaches. Granger causality, for example, emerges as a technique that can be better exploited for the analysis of this phenomenon.

As any other research, the present study has some limitations mostly related to the methodology adopted. The first limitation is related to the information source. Although the data was obtained from the Scopus database, a legitimate tool by the scientific community to conduct bibliometric research this database does not have complete quantitative coverage of publications. In thus vein, future studies may overcome this limitation by including additional databases (e.g., Web of Science). In relation to the composition of the unit of analysis, the dataset of this research consists only of articles. Future research should utilize more publication types, such as books, book chapters, conference papers, and editorials, to obtain a complementary picture of the knowledge development.

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